
A LIGHTWEIGHT CNN FOR DEEPPAKE CLASSIFICATION AND SEGMENTATION ON THE SID-SET

RESEARCH PAPER

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ABSTRACT

This paper presents a dual-pipeline framework for detecting and localizing image tampering on the Social media Image Detection dataset (SID-Set). The first pipeline is a ResNet-50-based classifier that predicts whether an image is *Real*, *Synthetic*, or *Tampered*. The second is a U-Net-style segmentation model with a ResNet-50 encoder that produces pixel-level masks of manipulated regions. On the official SID-Set test set, the classifier achieves a macro F1-score of 0.8651 across the three classes. The segmentation model attains a mean Dice score of 0.6309 and a mean Intersection-over-Union (IoU) of 0.5398 on tampered images. We demonstrate that our framework, despite being approximately $125\times$ smaller than the state-of-the-art SIDA-7B model (56M vs 7B parameters), achieves superior localization precision (IoU 0.5398 vs. 0.438). This establishes a strong, computationally efficient baseline for general-purpose deepfake analysis suitable for real-time deployment.

Keywords deepfake detection, image segmentation, U-Net, convolutional neural networks, image tamper localization

1 Introduction

The rise of generative models (Rombach et al., 2022) presents a challenge to digital authenticity. These models enable the creation of synthetic media which, beyond creative applications, pose risks of misuse. Such risks include misinformation campaigns and the erosion of trust in digital content. Consequently, distinguishing authentic images from synthetic or tampered content is a priority for digital forensics and information security.

Initial detection methods focused on facial manipulations (Rössler et al., 2019). However, current techniques generate full synthetic scenes or introduce object manipulations, demanding generalized analysis tools. The challenge extends beyond classification to localization, identifying manipulated pixels. Pixel-level segmentation is essential for forensic analysis and interpretable results.

To address this, our study proposes a dual-pipeline framework for image authenticity verification, evaluated on the Social media Image Detection dataset (SID-Set) (Huang et al., 2024). This benchmark contains 300,000 images categorized as real, AI-generated, and tampered, with pixel-level masks suitable for developing classification and segmentation models. Our framework decouples these tasks to optimize performance for each objective. We compare our approach with the SIDA-7B model to examine the necessity of large multimodal architectures for tamper localization.

Our contributions are threefold. First, we develop a classification model using a ResNet-50 backbone and Focal Loss to distinguish authentic, synthetic, and tampered images. Second, we design a segmentation model based on a U-Net with a ResNet-50 encoder, trained with a hybrid loss function to localize manipulated regions. Finally, we present an evaluation on the SID-Set benchmark, detailing training methodologies to establish a reproducible baseline.

2 Related Work

2.1 Detection and Localization Architectures

Deepfake detection is typically approached as a classification task. Convolutional Neural Networks (CNNs), such as ResNet He et al. (2016), are used to learn features that identify generative artifacts. For localization, which aims to create a binary mask of manipulated pixels, the U-Net architecture Ronneberger et al. (2015) is widely used. Its encoder-decoder structure with skip connections combines semantic context with high-resolution spatial details. Approaches often combine these architectures, using a pre-trained ResNet as the U-Net’s encoder to leverage pre-trained features for segmentation. The performance of these models also depends on the loss function; choices include Binary Cross-Entropy (BCE) for pixel-wise accuracy and the Dice coefficient Milletari et al. (2016) to handle class imbalance when manipulated regions are small.

2.2 Benchmark Datasets and Frameworks

The field has progressed from early datasets like CASIA Dong et al. (2013) and video-based benchmarks like FaceForensics++ Rössler et al. (2019), which focused on facial manipulations. However, many datasets lack the scale and diversity needed to address modern, general-purpose image tampering. The Social media Image Detection dataSet (SID-Set) Huang et al. (2024) marks an advance by providing a large-scale collection of images with manipulations commonly found on social media. It includes fully synthetic images and object-level tampering with corresponding masks, making it a testbed for generalized analysis. The dataset was introduced alongside the SIDA framework, which performs detection, localization, and provides textual explanations for its judgments.

3 Methodology

Our system is built around two dedicated pipelines: one focused on image classification and one on tampering segmentation. We designed them independently to keep the training process stable and the results easy to interpret. Although multi-task models with shared backbones are common, they can suffer from conflicting gradients that harm performance on one or both tasks. Using two specialized models avoids these issues and allows us to analyze and improve each task in isolation.

3.1 Dataset

All experiments use the SID-Set dataset Huang et al. (2024), which contains 300,000 images evenly divided into three classes: real, synthetic, and tampered. The official split provides 210,000 images for training, 30,000 for validation, and 60,000 for testing. Images in the tampered class include pixel-level binary masks indicating the manipulated regions, as shown in Figure 1.

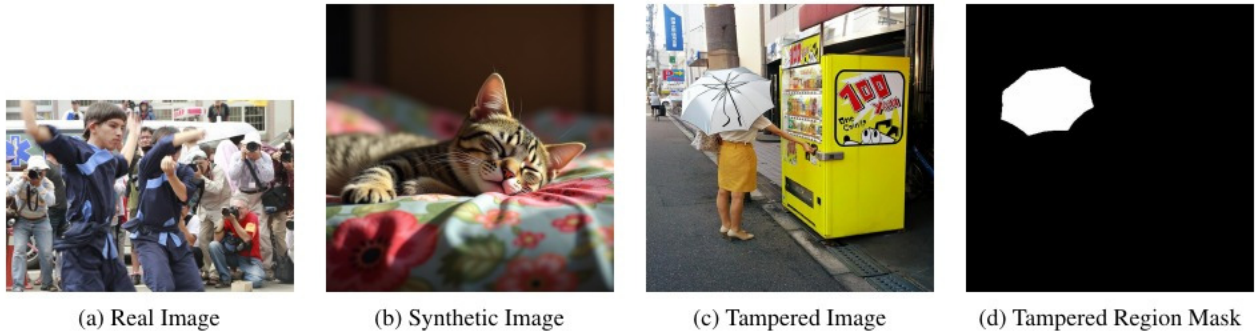


Figure 1: Samples from the SID-Set dataset: real, synthetic, tampered images, and a manipulation mask.

To enable model development with limited computational resources, our experiments used a fixed, representative subset of the data. For classification, we use the first 10,000 training and 3,000 validation images. For segmentation, we use the first 10,000 tampered images from the training set. Using approximately 4.7% of the available training data impacts generalization but allows for faster iteration during hyperparameter tuning. This approach ensures reproducibility while maintaining the original dataset’s class distribution. The classification model is trained to predict one of three labels,

while the segmentation model is trained exclusively on tampered images to localize manipulated regions using the ground-truth masks.

3.2 Pipeline A: Deepfake Classification

The classification pipeline was designed to distinguish images as Real, Synthetic, or Tampered. The following sections detail the rationale behind our configuration and training strategy.

3.2.1 Preprocessing

Input images were resized to a uniform 224x224 resolution. This specific size was chosen as it is the standard input dimension for ImageNet-pre-trained models Rosebrock (2017), including ResNet-50, ensuring that the feature distributions align with the backbone’s pre-trained weights. This choice provided a significant improvement in training efficiency with negligible impact on final accuracy during preliminary tests. The dataloader batch size was set to 64 to maximize GPU memory utilization.

3.2.2 Model Architecture

We adopt a ResNet-50 backbone He et al. (2016), as it provides a well-established balance between feature extraction capability and computational demand. The backbone is initialized with ImageNet-pretrained weights Deng et al. (2009) to leverage its learned representations.

During preliminary experiments, we found that unfreezing and training all layers from the start led to overfitting and disrupted the general-purpose features learned from ImageNet. To preserve robust low-level feature detectors, the initial layers of the backbone (conv1, bn1, and layer1) were frozen. This allows the deeper layers to adapt specifically to the distributional artifacts characteristic of manipulated or synthesized images.

As illustrated in Figure 2 the backbone outputs a 2048-dimensional global feature vector, which is passed to our custom classification head. The head consists of a dropout layer ($p = 0.3$), used to regularize the learned high-level features and improve generalization, followed by a fully connected linear layer that maps the feature vector to three output logits corresponding to the classes real, synthetic, and tampered.

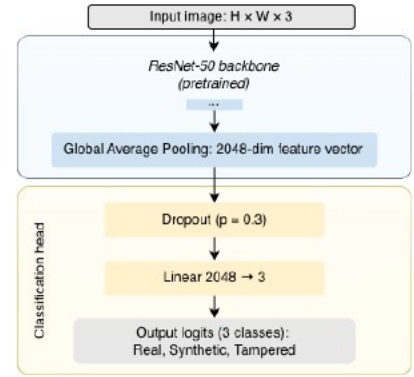


Figure 2: ResNet-50–based classification architecture.

3.2.3 Training Strategy

We design our training approach to address a key challenge: the three classes are not equally difficult to distinguish. Real and synthetic images are easier to separate, while tampered images present a significant obstacle.

When we begin with standard Cross-Entropy loss, we encounter a fundamental problem. The model picks up on real and synthetic images relatively quickly — those distinctions are fairly apparent. Tampered images, however, consistently get confused with real ones. The issue becomes clear: the loss is dominated by the two easier classes, spending most of its optimization effort on what the model already learns well. This leaves little capacity for the model to properly learn the subtle artifacts that define tampering.

Our approach leverages Focal Loss with $\gamma = 2.0$. Rather than treating all examples equally, Focal Loss down-weights the contribution from well-classified examples—the real and synthetic images—and concentrates the model’s learning capacity on the genuinely difficult tampered class. This ensures the model focuses on learning what actually distinguishes tampered images from real and synthetic ones.

To ensure stable convergence, we select the **AdamW** optimizer for its effective weight decay mechanism. We pair it with a **cosine annealing learning rate schedule**, which enables broad exploration of the loss landscape during early epochs before converging toward a robust minimum. For final evaluation, we employ an **Exponential Moving Average (EMA)** of the model’s weights. We observe that this time-averaged model consistently delivers better generalization on the validation set compared to the weights from any single epoch, as it effectively smooths out the stochasticity inherent in late-stage training.

3.3 Pipeline B: Deepfake Segmentation

The goal of this pipeline is to produce a pixel-level binary mask that localizes manipulated regions within a tampered image.

3.3.1 Model Architecture

For pixel-wise tampering localization, we adopt a U-Net architecture Ronneberger et al. (2015) that, similar to the classification pipeline, integrates a pretrained ResNet-50 as the encoder. The encoder comprises the ResNet stem (7×7 convolution, batch normalization, ReLU, and max pooling) followed by four residual stages (L1–L4), which progressively reduce spatial resolution while increasing feature depth. This structure provides multi-scale contextual representations essential for segmentation.

The decoder mirrors the encoder through four upsampling blocks. Each block upsamples the feature map by a factor of two and concatenates it with the corresponding encoder feature map via a skip connection. These skip connections are critical for restoring spatial detail lost during downsampling. Each decoder block applies convolution and dropout for regularization, and the channel dimensionality is gradually reduced (e.g., $2048+1024 \rightarrow 512$, $512+512 \rightarrow 256$, etc.). After the final upsampling stage, a 1×1 convolution produces the raw per-pixel logits, which are upsampled to the original input resolution. The full segmentation architecture is illustrated in Figure 3.

3.3.2 Data Preprocessing and Augmentation

Following the classification pipeline, we resize input images to 224×224 and normalize them using ImageNet statistics. For segmentation, a key challenge arises when augmenting images and masks: they must remain perfectly aligned. We address this by applying a joint horizontal flip to each image-mask pair, ensuring both transform identically. After this geometric augmentation, we apply image-only photometric augmentations—color jittering and Gaussian blur—to improve robustness without affecting the masks. Finally, we binarize the ground-truth masks to establish a clean, unambiguous training target.

A challenge in training the segmentation model is class imbalance in the dataset. As illustrated in Figure 4, the distribution of manipulated pixels is skewed: the median tampered area is 10.13%, and 10.0% of images have masks covering less than 1% of the area. This imbalance causes loss functions to become dominated by background pixels, resulting in poor localization of tampered regions.

To address this, we employ a hybrid loss function that combines two components. The total loss is a weighted sum of Binary Cross-Entropy with Logits (BCE) and soft Dice loss:

$$\mathcal{L}_{\text{total}} = 0.3 \cdot \mathcal{L}_{\text{BCE}} + 0.7 \cdot \mathcal{L}_{\text{Dice}} \quad (1)$$

BCE ensures per-pixel correctness, while Dice loss effectively handles class imbalance by focusing on region-based similarity. Initially, we train the model using only BCE loss. However, after qualitative analysis of predictions and further investigation into class imbalance in segmentation, we recognize that Dice loss is essential. Through additional training iterations, we discover that the Dice loss weight becomes increasingly important for the task. Consequently, we prioritize region-level tamper detection and class balance over strict per-pixel accuracy.

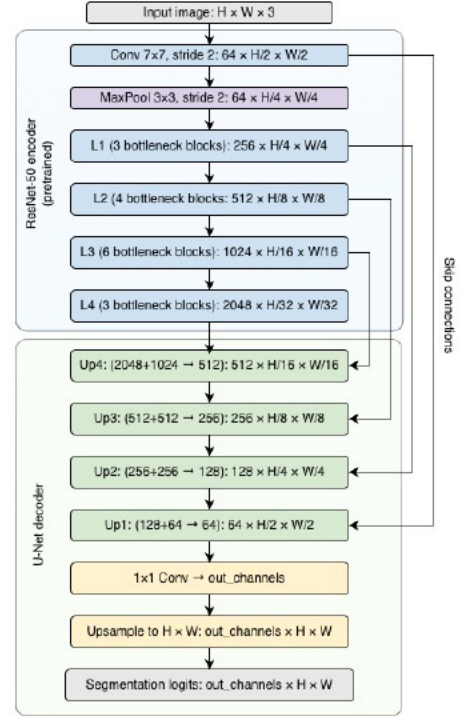


Figure 3: U-Net architecture

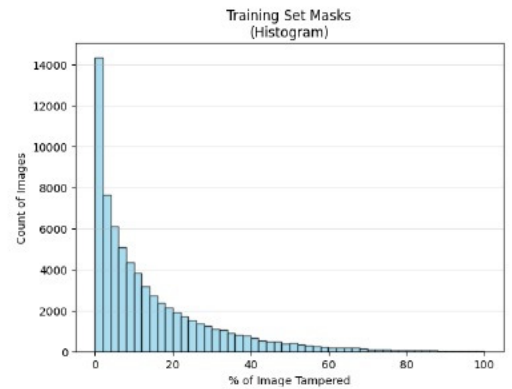


Figure 4: Distribution of tampered pixel percentages across the training set (median: 10.13%, with 10% of images having masks covering less than 1% of pixels).

The model is trained using the AdamW optimizer with a One-Cycle learning rate schedule Smith and Topin (2017), which systematically manages the learning rate to promote faster convergence. As with the classification model, we use an Exponential Moving Average (EMA) of the model’s weights for evaluation, accelerate training with Automatic Mixed Precision (AMP), and use gradient clipping to ensure stability.

3.4 Final Evaluation on the Official Test Set

We conduct a one-time evaluation on the official, held-out SID-Set test set to obtain an unbiased measure of final performance. After numerous iterations addressing challenges such as overfitting and training instability, we select our best-performing models based on peak validation metrics. This final test set comprises 60,000 unseen images sourced from the project’s GitHub and is used only after all model development and selection are complete.

For the classification task, we evaluate the EMA model checkpoint with the highest validation accuracy. For the segmentation task, we use the EMA model with the best validation Dice score to generate masks for the tampered test images. We measure final performance with a classification report and confusion matrix for classification, and mean Dice and IoU scores for segmentation. This final assessment provides the definitive measure of each model’s generalization capability.

4 Results

This section presents the performance of our classification and segmentation pipelines, beginning with an analysis of their training dynamics and validation performance before reporting final results on the official SID-Set test set. We conclude with a direct comparison to the state-of-the-art SIDA baseline and a discussion of our findings.

4.1 Pipeline A: Deepfake Classification

Our final classification model achieves a strong macro F1-score of 0.8651 on the official test set. The model excels at identifying synthetic images but, as anticipated, struggles to distinguish between real and subtly tampered images—the central challenge in this domain.

This behavior is detailed in the confusion matrices for both the validation and test sets (Figure 5). On the test set, the classifier identifies synthetic images with near-perfect accuracy (19,972 out of 20,000). However, a significant number of tampered images are misclassified as real, and vice-versa. This suggests the model struggles to spot the fine tampering details when the rest of the image looks normal.

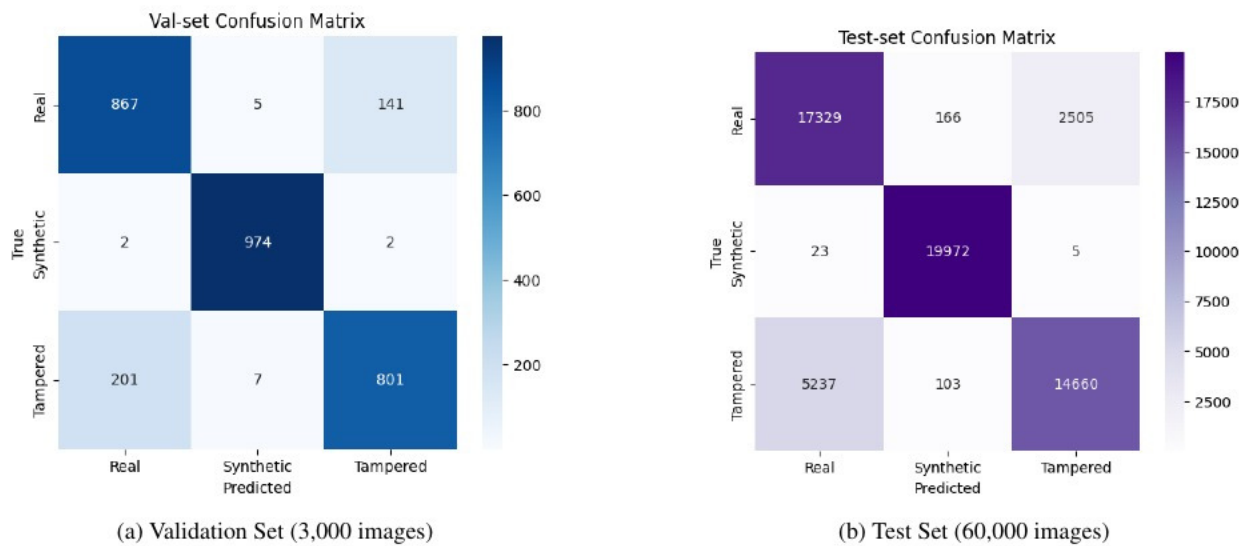


Figure 5: Confusion matrices for the classification model on (a) the validation subset (3,000 images) and (b) the official test set (60,000 images)

This pattern indicates that the model struggles to detect subtle manipulations that leave global semantics largely unchanged, which is the primary challenge for general-purpose deepfake classification Gupta et al. (2025).

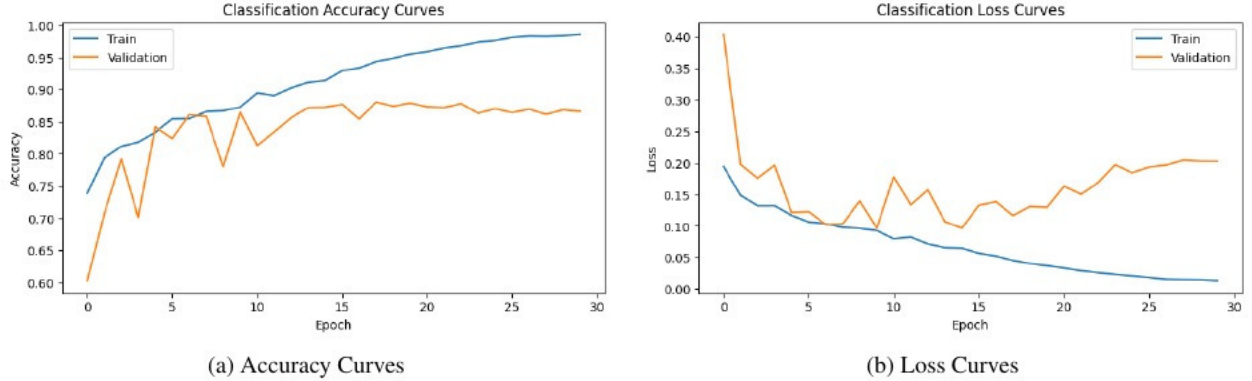


Figure 6: Training and validation curves for the classification model over 30 epochs. The model shows good convergence but also a clear generalization gap, as validation performance plateaus while training performance continues to improve.

The training curves in Figure 6 illustrate the model’s learning dynamics. The training accuracy converges to 98.6%, while the validation accuracy plateaus around 88.1%. This gap suggests a degree of overfitting to the training subset, a common challenge that could be addressed with a larger training dataset or more aggressive regularization. The validation loss curve bottoms out around epoch 15 before slightly increasing, which confirms this trend. The final model checkpoint was selected from the epoch with the highest validation accuracy (Epoch 18, 88.1%).

4.2 Pipeline B: Deepfake Segmentation

The segmentation model obtains a mean Dice score of 0.6309 and a mean IoU of 0.5398 on tampered images in the official test set (Table 1).

These results indicate a reasonable ability to localize manipulated regions, while leaving substantial room for improvement, especially given the strong class imbalance in mask coverage.

Table 1: Summary of main results on the SID-Set test set.

Model	Task	Metric (Value)
Classification	Real/Synthetic/Tampered	Macro F1: 0.8651
Segmentation	Tamper Localization	Mean Dice: 0.6309 Mean IoU: 0.5398

The training process, visualized in Figure 8b, shows the training Dice score steadily improving to approximately 0.80. The validation Dice score, while more volatile, peaks and stabilizes around 0.70. This gap between training and validation performance again suggests overfitting. The best model checkpoint was saved based on the highest validation Dice score achieved during training.

Further analysis on the test set involved a probability threshold sweep, shown in Figure 8a. The plot illustrates the trade-off between precision and recall, as the threshold increases, precision rises while recall falls. The Dice score, which balances these two metrics, peaks at approximately **0.78** when the probability threshold is set to around 0.63. This suggests that for practical applications, a threshold higher than the default 0.5 produces more balanced and accurate segmentation results.

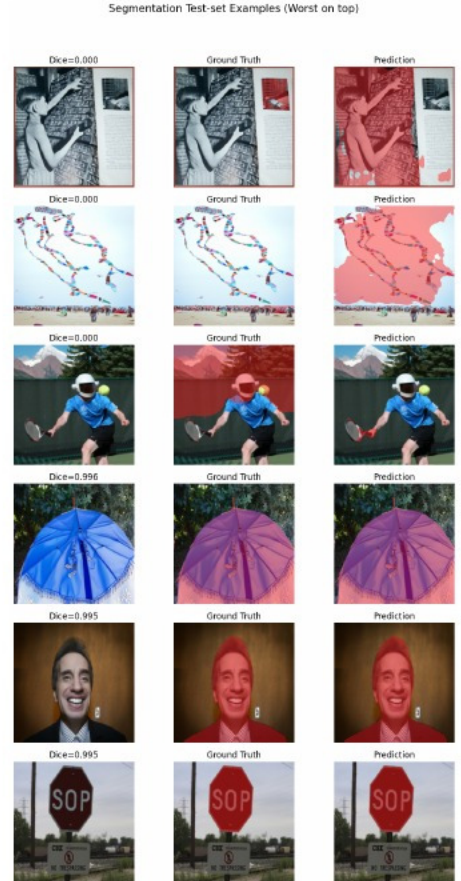


Figure 7: Showcasing the three best and three worst validation masks.

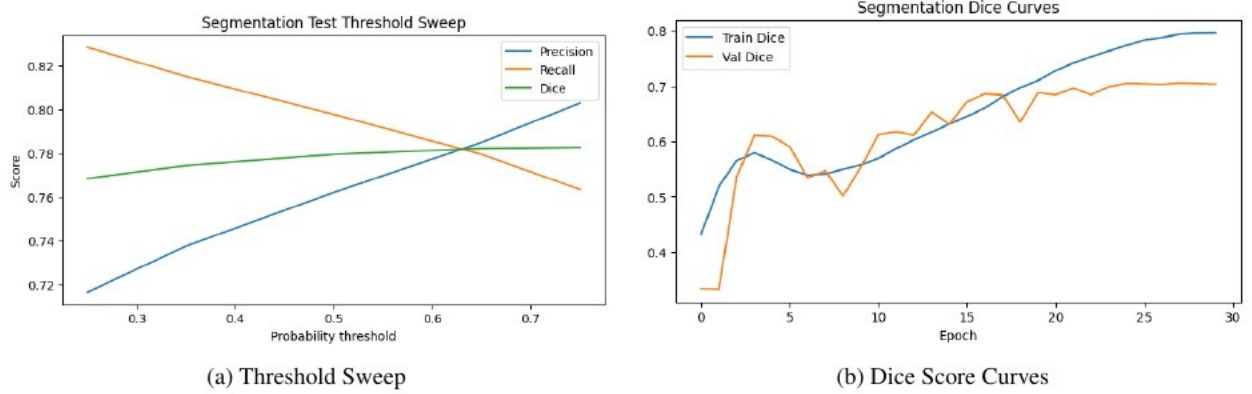


Figure 8: Performance of the segmentation model. The left plot shows the Dice score on the test set across different probability thresholds, peaking near 0.78. The right plot shows training and validation Dice score curves, indicating a generalization gap.

4.3 Comparison with the Official Benchmark Baseline

The SID-Set benchmark was introduced alongside the SIDA framework Huang et al. (2024), establishing SIDA-7B as the primary reference model for this dataset. Our evaluation focuses on this comparison to determine if standard, computation-efficient architectures can compete with the massive multimodal models proposed by the benchmark authors.

On the SID-Set test split, SIDA-7B reports an overall detection F1-score of 93.5%. Our ResNet-50 classifier achieves a macro F1-score of 0.8651. This performance gap is expected given SIDA’s advantages: it utilizes a large vision–language model, higher input resolutions (1024×1024), and training on the full dataset.

However, for localization on tampered images, our specialized model demonstrates superior precision. SIDA-7B attains an IoU of 43.8. Our U-Net baseline, despite being trained on only a subset, reaches a mean IoU of 53.98.

This result is significant. While specialized forensic architectures exist, our findings demonstrate that even a standard U-Net outperforms the state-of-the-art multimodal baseline on this dataset Dong et al. (2023). This suggests that for pixel-level localization, the massive parameter count of Vision-Language models does not necessarily translate to higher precision, and that efficient, supervised baselines remain highly competitive.

4.4 Computational Efficiency

A distinct advantage of our dual-pipeline framework is computational efficiency. As shown in Table 2, our combined framework uses approximately **56 million parameters** (23.5M for classification, 32.5M for segmentation). This represents less than 1% of the 7 billion parameters required by the SIDA-7B model. Our use of standard CNN backbones and 224×224 resolution enables high-throughput inference (>100 FPS on standard GPUs), making it suitable for real-time filtering. Large multimodal models, by contrast, typically require significant VRAM and computational overhead.

Table 2: Computational efficiency benchmarks on an NVIDIA RTX 5060 Ti. SIDA-7B inference speed was not measured in this study, but its parameter count is provided for scale comparison.

Model	Parameters	Throughput	Latency	IoU
SIDA-7B (Huang et al., 2024)	7,000 M	—	—	43.8
Ours (Classifier)	23.5 M	234 FPS	4.28 ms	-
Ours (Segmentor)	32.5 M	169 FPS	5.93 ms	53.98
Ours (Combined)	≈ 56 M	98 FPS	10.21 ms	53.98

4.5 Discussion

The results highlight the capabilities and limitations of convolutional architectures on SID-Set. The classifier identifies synthetic images with high accuracy, indicating that generative artifacts remain detectable in social media settings.

However, confusion between real and tampered images reveals two design constraints. First, resizing images to 224×224 removes high-frequency artifacts—noise patterns and edge inconsistencies—needed to detect local tampering. Second, the gap between training and validation performance suggests the 10,000-image subset is insufficient to capture the variance in manipulation techniques. Increasing input resolution and training set size would likely improve the distinction between authentic and tampered content.

For localization, our model achieves a 53.98 mean IoU compared to SIDA’s 43.8, indicating a trade-off in architecture. While SIDA-7B uses a Vision-Language approach to ‘reason’ about tampering, the tokenization process in Large Multimodal Models often degrades spatial information. In contrast, the U-Net retains spatial feature maps via skip connections. This suggests that while Multimodal models excel at semantic detection (identifying *what* is fake), convolutional networks remain effective for precise localization (identifying *where* is fake).

Currently, we evaluate the two pipelines independently. In a practical workflow, a unified system would first determine if an image is tampered, then localize manipulations.

5 Conclusion and Future Directions

We introduced a dual-pipeline framework for deepfake analysis on the SID-Set benchmark, using a ResNet-50 classifier for detection and a U-Net with a ResNet-50 encoder for localization. By documenting our methodology, we provide a reproducible baseline for image tampering detection and localization.

Our classifier achieved a 0.8651 macro F1-score on the SID-Set test set. While synthetic images were identified accurately, errors occurred primarily in distinguishing real from tampered content. A generalization gap suggests the model would benefit from more data. For segmentation, the model obtained a 0.6309 mean Dice and 0.5398 mean IoU. Adjusting the decision threshold from 0.5 to 0.65 improved the Dice score to 0.78, highlighting the importance of threshold selection.

Compared to SIDA, our model has a lower detection F1-score but achieves competitive localization performance with significantly less computational cost. This reinforces the value of supervised CNN baselines.

Future work should address the resource constraints identified in this study. Specifically, training on the full 210,000-image dataset would reduce overfitting, and increasing input resolution (e.g., to 512×512) would help preserve subtle manipulation traces. Further research could explore multi-task architectures with shared backbones or transformer-based encoders to improve efficiency. Additionally, robustness analysis against compression and resizing is needed. Finally, these components could be integrated into an end-to-end pipeline where localization is triggered by the classifier’s output.

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